ASHISH KOSANA

Data Analyst • Data Engineer • ML Engineer

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PROFESSIONAL SUMMARY

Results-driven data professional with a B.S. in Computer Science (GPA 3.51, Dean's List) and hands-on experience across the full data lifecycle — from pipeline engineering and data modeling to machine learning, statistical analysis, and executive-ready visualization. Proficient in Python, SQL, R, Tableau, Power BI, Snowflake, AWS, and Azure. Experienced building ETL pipelines, dashboards, and ML systems deployed on cloud infrastructure. Skilled at translating complex datasets into actionable business insights and communicating findings clearly to both technical and non-technical stakeholders. Holds Microsoft Certified: Azure Database Administrator Associate (DP-300). Currently authorized to work in the U.S. on OPT (F-1); open to H-1B sponsorship and relocation.

TECHNICAL SKILLS

Languages & Scripting: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL (CTEs, window functions, DDL), R, Bash, Scala

Data Engineering & ETL: ETL Pipeline Design, Snowflake, Apache Spark, Hive, Hadoop, Informatica, SSIS, AWS Glue, AWS Lambda, S3, Docker

BI & Visualization: Tableau (LOD, parameters, Tableau Cloud), Power BI (DAX, star schema, RLS, time intelligence), Matplotlib, Seaborn, ggplot2, Excel

Machine Learning & AI: Supervised Learning, Classification, Regression, ANN, RNN, LSTM, GRU, NLP, Feature Engineering, MLflow, Hyperparameter Tuning

Cloud & DevOps: AWS (EC2, ECR, Elastic Beanstalk, Glue, Lambda, S3, KMS), Azure, GCP, Databricks, Git, CI/CD, Streamlit, Jupyter Notebooks

Data Governance: Data Quality Validation, Metadata Management, Access Controls, Row-Level Security, OWASP, NIST SP 800-160, Data Lineage

PROFESSIONAL EXPERIENCE

Data & Operations Analyst • Aramark • UMass Lowell, MA Sep 2024 – Dec 2025

- Cleaned and validated 500+ weekly transaction records using Python and SQL, reducing data errors and improving downstream reporting accuracy for operational and management teams.
- Built and maintained Power BI dashboards integrating Excel datasets, increasing frontline management visibility into operational performance metrics.
- Automated data entry and validation workflows with Python scripts, cutting manual processing time and standardizing data pipelines across shift reporting cycles.
- Documented data discrepancies and produced structured resolution reports, clearly communicating data quality issues to non-technical supervisors.
- Standardized daily sales data extraction using SQL queries, supporting consistent operational tracking across 4 daily shifts and reducing reporting inconsistencies.

PROJECT EXPERIENCE

Sales Performance Analytics Platform • Independent Project 2025

- Architected star-schema data model in Power BI with 15+ DAX measures including YoY growth, rolling averages, time-series forecasting, and margin decomposition supporting multi-region business performance analysis.
- Implemented what-if scenario parameters for revenue modeling under variable pricing assumptions and deployed row-level security for regional stakeholder access control.
- Partnered with stakeholders to gather analytical requirements and delivered self-service dashboards that reduced ad-hoc reporting requests from business development teams.

AWS Usage & Cost Intelligence Dashboard (500K+ rows, Snowflake, Tableau) • Independent Project 2025

- Ingested and transformed 10M+ row AWS usage datasets into Snowflake using optimized SQL (CTEs, window functions, clustering keys), building a clean enterprise data pipeline for downstream analytics.

- Built multi-layer Tableau dashboards surfacing cost allocation trends and anomalies for finance and engineering teams, enabling action within reporting cycles.
- Demonstrated measurable compute cost reduction through pre/post benchmarking queries; produced executive presentation materials translating pipeline findings into financial impact narratives.

Student Performance Indicator — End-to-End ML Pipeline (AUC 0.91) • Independent Project2025

- Engineered full data ingestion, validation, and transformation pipeline using Python and Scikit-learn; implemented MLflow experiment tracking for hyperparameter logging and reproducibility documentation.
- Deployed containerized ML application on AWS (EC2, ECR, Elastic Beanstalk) using Docker with cross-cloud portability on Azure; built Jupyter notebook evaluation artifacts for stakeholder review.
- Addressed class imbalance using SMOTE and undersampling techniques; computed and documented accuracy, precision, recall, F1, and AUC-ROC metrics across cross-validation runs.

Student Mental Health & Depression Analysis (AUC 0.88, 5,000 students) • Independent Project2024

- Designed SQL Server data pipeline for multi-source extraction, transformation, and multivariable statistical correlation modeling across academic and lifestyle factors.
- Published interactive Tableau Cloud dashboard with parameter-driven demographic filtering, enabling non-technical stakeholders to independently explore population-level risk patterns.
- Applied R (ggplot2) for exploratory analysis and cross-validated findings against Python outputs before final reporting.

Bank Churn Prediction — ANN Classification System (AUC 0.83) • Independent Project 2025

- Developed ANN churn classification model using Python, Scikit-learn, and TensorFlow with preprocessing pipelines for feature scaling, engineering, and train/validation/test split management.
- Integrated trained model into Streamlit inference application with real-time prediction and payload schema validation; delivered model evaluation documentation in Jupyter notebooks.

Next Word Prediction — LSTM & GRU Language Model • Independent Project 2025

- Designed LSTM and GRU sequence prediction models in Python and TensorFlow with end-to-end text preprocessing pipelines and iterative hyperparameter tuning cycles.
- Built interactive Streamlit inference interface and Jupyter evaluation notebooks documenting model architecture decisions, training run logs, and performance metrics.

Movie Review Sentiment Analysis — RNN NLP Pipeline • Independent Project 2025

- Built RNN-based NLP sentiment classification pipeline in Python and TensorFlow with text preprocessing, tokenization, and feature engineering for sequence modeling on the IMDB dataset.
- Deployed Streamlit application with real-time sentiment prediction; maintained Git version-controlled codebase with full reproducibility documentation.

EDUCATION

B.S. Computer Science • GPA: 3.51 • Dean's List • University of Massachusetts LowellSep 2024 – Dec 2025

B.Tech. Computer Science • GPA: 7.5 / 10 • SR International Institute of Technology, Telangana, India Oct 2021 – Apr 2024

CERTIFICATIONS

- Microsoft Certified: Azure Database Administrator Associate (DP-300)
- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp — Udemy (2025)
- Complete Data Analyst Bootcamp: Basics to Advanced — Udemy (2025)
- Career Essentials in Data Analysis — Microsoft & LinkedIn Learning
- Data Analysis with Python — freeCodeCamp